T-201-GSKI Data Structures

Programming Homework 3: Binary Search Trees (5%)

Objective

The objective of the assignment is to gain hands-on experience implementing binary search trees, as well as exploring interesting use cases for them.

Logistics

- You are expected to work on the assignment in a ***group of 1-3 persons*** (max).
- *Ensure that you test your implementations thoroughly* (it is easy to make mistakes or miss some important edge cases).
- Do not change the names of the provided Python files; they must remain as is.
- ***Submit your code in Canvas***. Archive the folder where you keep your Python files to create either as a *zip* or a *tar/tgz* file. Submit only that single archived file.

Description

You are provided with a based codebase that you will work with. You are *not allowed* to change the interface of the provided class methods (you can add *private* methods, if you want). Implement the missing parts (marked 'To do').

1. (70%) Implement the incomplete methods (marked 'To do') of the *BinarySearchTree* class (in binary_search_tree.py).

   IMPORTANT: Ensure that your ***insert*** method is working properly before starting to implement other methods. Otherwise, it will be difficult (for both you and us) to test your code. Submissions with a non-working insert will receive 0 marks for part 1.

2. (15%) Only start this part once your BST implementation from above is (mostly) finished and has passed basic tests. Otherwise, this part cannot be tested nor graded.
   You are going to use your BST as an underlying data type for implementing a Python like dictionary. For that, complete the incomplete methods of the *MyDict* class (*my_dict.py*), marked 'To do'.

3. (15%) Only start this part once your dictionary implementation from above is finished and has passed some basic tests. Otherwise, this part cannot be tested nor graded.
   In here you will showcase your dictionary implementation, by using it to finish the task(s) set out in *work_freq.py*.

Good luck! 😀